Computer Vision Syndrome Smart (CVS-Smart) Questionnaire and Score

CVS-Smart questionnaire[1, 2]

The Computer Vision Syndrome–Smart (CVS-Smart) questionnaire is a novel reliable and valid ultra-short self-assessment questionnaire that can definitely diagnose CVS cases without the need of ophthalmic examination.[1] The CVS-Smart is linked to an additional score (CVS-Smart score) that provides the suitable diagnosis and recommendation.[1, 2] CVS-Smart is supported by Iqbal's anti-CVS protective measures (Iqbal's instructions) to re-schedule the screen time and style to guard against the potential CVS sequelae.[1, 2] The CVS-Smart included only five questions that were developed from the reliable and valid CVS-F4 questionnaire[1, 3] (Kuder Richardson 20 Formula .81 and Content Validity Index 0.92)[1] and the experts' recommendations.[1, 3, 4] CVS-Smart questionnaire is ideal for University students and young adults.[1] Both CVS-Smart questionnaire and score are available as online tool at SurveyMonkey web site.[2] However, it could be printed in a paper format as desired for personal use or research work.

CVS-Smart online link2

https://www.surveymonkey.com/r/YHXVNWC

Informed/Electronic consent[1, 2]

Please note that your response to the *online CVS-Smart* is considered as an informed or *electronic consent* that we could use the resultant data in the scientific research and analysis to improve the CVS-Smart and score. Be sure that there is no personal data required in this questionnaire and your response is anonymous (no names required for online CVS-Smart).

Alternatively, you can simply print the CVS-Smart questionnaire and score for personal use to self-assess your CVS-status. Individual/Personal use of CVS-Smart requires no permissions. Permissions are only required when CVS-Smart is used for research work.

Reliability and validity of CVS-Smart questionnaire[1, 2]

Variables	CVS-Smart
Number of questionnaire items	5
Responses	3 responses/answers/choices for each questionnaire item
Reliability indices:	
Cronbach's Alpha Coefficient	.860 (high reliability)
Guttman Split-Half Coefficient	.805 (high reliability)
Interrater Reliability (Spearman's rho correlation coefficient):	
-Significance at a 0.01 significance level	significant for all questionnaire items
-Correlation Interpretation	Moderate to strong correlation for all questionnaire items (range .511-.691)
Validity indices:	
I.Content validity (by 5 experts)	
-Content Validity Ratio (CVR)	1 for each of 5 questionnaire items
-Content Validity Index (CVI)	1 (perfect content validity)
II.Face validity (by non-experts) Face validity was evaluated by some medical students (participants)	Strong, clear, relevant, appropriate, concise, simple, easy to understand, comprehensive and all 5 items are of same importance
III.Construct validity	
A.Confirmatory Factor Analysis (CFA)	
-Comparative Fit Index (CFI)	.986 (good fit)
-Tucker Lewis Index (TLI)	.971 (good fit)
-Root Mean Square Error of Approximation (RMSEA)	.078 (mediocre fit)
B.Spearman's rho correlation coefficient	
-Significance at a 0.01 significance level	significant for all questionnaire items
-Correlation Interpretation	strong correlation for all questionnaire items (range .771-.830)

How to self-assess yourself

Please, read and respond carefully to the following ***5 questions***, then refer to the CVS-Smart score to diagnose your CVS-status by yourself and discover a friendly-guide tool to manage your case. Both CVS-Smart score and Iqbal's instructions are linked at the end of this CVS-Smart questionnaire. CVS-Smart score of 7-10 score points signifies a positive CVS-case.[1, 2]

Computer vision syndrome-Smart questionnaire (*CVS-Smart*) [1, 2]

Name (optional): **Age:** **Sex:** **Occupation:**

5 Questions	Content	CVS diagnostic criteria	Score
1. Visual	**9 visual complaints:** Blurred vision Eye strain/fatigue Double vision/diplopia Difficulty in refocusing the eyes Near vision discomfort/difficulty Unclear objects post-screen use Glare/seeing halos of light Feeling diminution of vision Increased sensitivity to light	***Question 1:*** **Out of these 9 visual complaints; how many complaints you suffer from?**	
		No visual complaints	**0**
		One visual complaint	**1**
		Two or more visual complaints	**2**
2. Ocular surface	**9 ocular surface complaints:** Dry eyes Eye redness Eye irritation/discomfort Foreign body sensation Burning sensation/stinging sensation Itching/eye rubbing Lacrimation/watery eye Eyelids feel heavy Frequent blinking	***Question 2:*** **Out of these 9 ocular surface complaints; how many complaints you suffer from?**	
		No ocular surface complaints	**0**
		One ocular surface complaint	**1**
		Two or more ocular surface complaints	**2**
3. Extraocular	**9 extraocular complaints:** Headache Neck pain Shoulder pain Back pain Joint pain in fingers and wrists Inability to hold objects well Difficulty to write using a pen Sleep disturbances/ insomnia Inattention/depression	***Question 3:*** **Out of these 9 extraocular complaints; how many complaints you suffer from?**	
		No extraocular complaints	**0**
		One extraocular complaint	**1**
		Two or more extraocular complaints	**2**
4. Frequency	Frequency of the complaints	***Question 4:*** **How do you describe the frequency of your complaints?**	
		Rare	**0**
		Infrequent	**1**
		Frequent	**2**
5. Screen-associated	The complaints should be related to and associated with the time of screen use	***Question 5:*** **Are your complaints associated with the time of the screen use? (i.e. are your complaints exist during the time of screen use?)**	
		Never	**0**
		Sometimes	**1**
		Always	**2**
Total score			**0-10 points**

The CVS-Smart Score[1, 2]

CVS-Smart total score	CVS-case probability status	Recommendations
7-10 points	Positive CVS-case *(CVS diagnosis is confirmed)*	*-Consult your ophthalmologist or optometrist to receive the appropriate treatment* *-Reduce your screen-time* *-Follow Iqbal's instructions* *-If you already have a chronic eye disease or pathology, or previous eye surgery; this diagnosis might be inaccurate in your condition*
5-6 points	High probability	*-Consult your ophthalmologist or optometrist to confirm or exclude CVS diagnosis* *-In case you are diagnosed as a positive CVS-case, please receive the appropriate treatment* *-If you are diagnosed as a negative CVS-case, please repeat CVS-Smart every 6 months for check-up* *-Reduce your screen-time* *-Follow Iqbal's instructions*
3-4 points	Low probability	*-Repeat CVS-Smart every 6 months for check-up* *-Reduce your screen-time* *-Follow Iqbal's instructions*
1-2 points	Not CVS-case	*-Repeat CVS-Smart every 6 months for check-up* *-Follow Iqbal's instructions to improve your score to 0 points*
0 points	Normal subject	*-Follow Iqbal's instructions as prophylactic measures*

Modified Iqbal's instructions[1, 2, 5]

Iqbal's anti-CVS protective measures		
Screen-time	**Screen-style**	**Eye-style and life-style**
-Reduce your screen-time to minimum *-1 screen-hour daily is ideal* *-Try not to exceed 2 screen-hours daily* *-Focus your screen-hours to accomplish your necessary assignments* *-Specify one place only e.g. home or college to accomplish your assignments*	*-Shift from hand-held to non-hand-held screens* *-Shift from small-sized to large-sized screens* *-Shift from old to new screen versions* *-Shift from screens to books for your studies* *-Print the necessary PDF or word files for your studies* *-Use speech language writing instead of keyboard*	*-Wear your spectacles during watching screens* *-Avoid/limit contact lens wearing* *-Treat your eye disease as dry eye with necessary medication*
-Desktop computer or laptop is preferred *-Avoid using smartphone to accomplish your assignments* *-Day screen-hours are more preferred than night screen-hours*	*-Shift from other screens to TV screens to attend/watch necessary online courses, videos, meetings, lectures and educational programs* *-Connect TV screens to internet and watch your desired movies, series and applications* *-The proper distance for watching TV is 4-6 meters*	*-Avoid taking your smartphone to bed at night when you are going to sleep i.e. keep your smartphone away from you when you are sleeping*
-Avoid or limit video-calls *-Use regular audio-calls* *-Try to limit smartphone to audio-calls and use other screens for video-calls if necessary* *-Uninstall the unnecessary applications in your smartphone and other screens* *-Minimize your social-media applications' time*	*-Avoid the following risk-factors that aggravate your subjective complaints:* *Poor screen-resolution or design* *Screen-glare* *Screen edge at/above horizontal eye level* *Close eye-screen distance* *Poor lighting conditions* *Watch screen in the dark* *Uncomfortable seating postures* *Small-font size* *Texting with both thumbs*	*-Specify more time for sports* *-Try to lose weight if you are overweight* *-Try to improve your social life, family relationships and gain more friends* *-Get more interest with your non-screen hobbies e.g. diving* *-Join society organizations, public services or community groups e.g. travelers' groups*
-Avoid continues screen watching, take breaks and you could apply to the 20-20-20 rule	*-Adjust screen brightness to minimum possible* *-Try not to exceed 50% screen brightness* *-Use smartphone holder instead of your hands*	*-Adjust the indoor and outdoor light source to be above or behind your eyes*

Permissions

To use CVS-Smart as a subjective instrument in research work, permissions should be requested from *Professor Mohammed Iqbal, MD, PhD*

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References

1. Iqbal M, Elmassry A, Elgharieb M, Said O, Saeed A, Ibrahim T, Kotb A, Abdelhalim M, Shoughy S, Elgazzar A, Shamselden H, Hammour A, Eid M, Elzembely H, Abdelaziz K. Visual, ocular surface, and extraocular diagnostic criteria for determining the prevalence of computer vision syndrome: a cross-sectional smart-survey-based study. Med Hypothesis Discov Innov Ophthalmol. 2024 Jul 1; 13(1): 1-15. https://doi.org/10.51329/mehdiophthal1489
2. CVS-Smart questionnaire (2021). 'Computer Vision Syndrome Smart (CVS-Smart) Questionnaire and Score', SurveyMonkey website. Available at: https://www.surveymonkey.com/r/YHXVNWC
3. CVS-F4 questionnaire (2021). 'Computer Vision Syndrome Form 4 (CVS-F4) Survey', SurveyMonkey website. Available at: https://www.surveymonkey.com/r/BTLGFDD
4. Iqbal M. Prevalence and Sequelae of Computer Vision Syndrome in Egyptian Universities. 2021. [Accessed: May 30, 2024]. Available at: https://clinicaltrials.gov/study/NCT05187221?term=NCT05187221&rank=1.
5. Iqbal M, Soliman A, Ibrahim O, Gad A. Analysis of the Outcomes of the Screen-Time Reduction in Computer Vision Syndrome: A Cohort Comparative Study. Clin Ophthalmol. 2023;17:123–134. doi: 10.2147/OPTH.S399044.